



MCIIS Theses and Capstone Repository

IT 228 – Advanced Information Management

By

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Administrator User Views of MCIIS Repository

Data Requirements:

User

Administrators manage user accounts, assign roles, and generate reports on user role distribution. The User entity includes attributes userID (unique identifier), studentID (nullable for non-student roles), firstName, middleName, lastName, contactNumber, email (valid USeP email and must be unique), role ("Administrator," "MCIIS Staff," "Faculty," or "Student"), password, and createdTimestamp.

Research

Administrators can view research information and generate reports on research access statistics and keyword usage. The Research entity includes researchID (unique identifier), uploadedBy (referencing the User entity), researchTitle (must be unique), researchAdviser (referencing the Faculty entity), program ("Bachelor of Science in Information Technology," "Bachelor of Science in Computer Science," "Bachelor of Library and Information Science," "Master of Library and Information Science," or "Master in Information Technology"), publishedMonth, publishedYear, researchAbstract (text), researchApprovalSheet (image), and researchManuscript (pdf file). They can also filter it by researchAdviser, program, and year.

Faculty

Administrators can add, edit, and delete the faculty list. The Faculty entity includes facultyID (unique identifier), firstName, middleName, lastName, position, designation, email (valid USeP email and must be unique), ORCID, contactNumber, educationalAttainment, fieldOfSpecialization, and researchInterest.

Researcher

Administrators can view researcher information. The Researcher entity includes researcherID, researchID (referencing the Research entity), firstName, middleName, lastName, and email (valid USeP email and must be unique).

Keyword

Administrators can view keywords. The Keyword entity includes keywordID (unique identifier) and keywordName (unique).

Panel

Administrators can view the list of faculty panelists to research projects. A faculty member can be a panelist for multiple research projects, while a research project can have multiple panelists. The Panel entity is a join table that handles the many-to-many relationship between faculty members and research entries that includes panelID (a unique identifier), facultyID (foreign key referencing the Faculty entity) and researchID (foreign key referencing the Research entity).

User Audit Log

Administrators are responsible for maintaining logs of profile-related activities to ensure accountability and security. The User Audit Log entity tracks any changes made to user or faculty profiles. It records details such as auditLogID (a unique identifier), modifiedBy (referencing the User entity, representing the administrator or faculty who made the change), targetUserID (referencing the User entity if a user profile was modified), actionType (e.g., "update user"), and timestamp (the date and time the change occurred).

Faculty Audit Log

Administrators are responsible for maintaining logs of profile-related activities to ensure accountability and security. The Faculty Audit Log entity tracks any changes made to user or faculty profiles. It records details such as auditLogID (a unique identifier), modifiedBy (referencing the User entity, representing the administrator or faculty who made the change), targetFacultyID (referencing the Faculty entity if a faculty profile was modified), actionType (e.g., "update faculty"), and timestamp (the date and time the change occurred).

ResearchAccessLog

Administrators can generate reports on the most research accessed entries. The Research Access Log entity includes accessLogID, researchID (referencing the Research entity), userID (referencing the User entity), and accessTimestamp.

KeywordSearchLog

Administrators use keyword search log to generate reports on the most frequently searched keyword. The Keyword Search Log entity includes searchLogID, keywordID (referencing the Keyword entity), userID (referencing the User entity), and searchTimestamp.

ResearchEntryLog

Administrators can view research entry logs to monitor repository activity and trace actions made on research entries. The Research Entry Log entity includes entryLogID (a unique

identifier), performedBy (referencing the User entity), actionType (create or modify), researchID (referencing the Research entity), and timestamp (indicating the date and time of the upload).

CompiledReport

Administrators can generate and download compiled reports to consolidate abstracts, executive summaries, and matrix or tabular reports based on various filters such as program, year, and adviser. The Compiled Report entity is designed to keep a record of these generated outputs for documentation and reference. It includes the following attributes: reportID (a unique identifier), reportType (the type of report, e.g., Abstracts, Executive Summaries, Statistics, or Matrix), generatedBy (foreign key referencing the User entity that indicates who generated the report), generatedOn (the date and time the report was created), filtersApplied (a JSON field containing the applied filters such as program, year, and adviser), and filePath (the storage location of the generated Word or PDF file that supports OCR for searchability).

Transaction Requirements:

Data Entry

- Add the details of a new user account, assigning an initial user role (Administrator, MCIIS Staff, Faculty, or Student).
- Add new faculty records with a complete profile.

Data Update/Deletion

- Update/delete the details of a user account.
- Update/delete faculty profiles.

Data Queries

- Search research by title or keyword.
- Filter research by adviser, program, or year.
- View profile information of faculty members.
- Generate reports on user role distribution (number of Administrators, MCIIS Staff, Faculty, and Student).
- Generate and download compilations of abstracts or executive summaries by Program, Year, and Faculty/Adviser, which include the research title, list of researchers, research adviser, month and year of publication, the abstract or executive summary, and associated keywords.
- Generate and download reports in matrix or tabular format (PDF/Excel) that include the following fields: Research Title, Date Completed (Month and Year), Researchers, Research Adviser, Keywords, Panel, Related Agenda, Related SDG, and Related SRIG.

- Identify the top 5 most accessed research entries based on access count.
- Identify the top 5 most frequently used keywords.
- View research counts per program.
- View research counts per year.

MCIIS Staff User Views of MCIIS Repository

Data Requirements:

Research

MCIIS Staff are responsible for adding and editing research metadata, ensuring all research entries are accurate, complete, and properly categorized. The Research entity includes researchID (unique identifier), uploadedBy (referencing the User entity), researchTitle (must be unique), researchAdviser (referencing the Faculty entity), program (“Bachelor of Science in Information Technology,” “Bachelor of Science in Computer Science,” “Bachelor of Library and Information Science,” “Master of Library and Information Science,” or “Master in Information Technology”), publishedMonth, publishedYear, researchAbstract (text), researchApprovalSheet (image), and researchManuscript (pdf file). They can also filter it by researchAdviser, program, and year.

Thematic tagging: Each research entry can be associated with one or more Agenda, SDG, and SRIG categories to enable classification, reporting, and dashboard analytics.

Faculty

MCIIS Staff can view the faculty list. The Faculty entity includes facultyID (unique identifier), firstName, middleName, lastName, position, designation, email (valid USeP email and must be unique), ORCID, contactNumber, educationalAttainment, fieldOfSpecialization, and researchInterest. They can also assign research to faculties who are research advisers and panelists, as well as generate productivity reports identifying the number of research entries advised and paneled per faculty member.

Researcher

MCIIS Staff can add, edit, and delete researcher information and ensure that each research entry has properly recorded contributors. The Researcher entity includes researcherID, researchID (referencing the Research entity), firstName, middleName, lastName, and email (valid USeP email and must be unique).

Keyword

MCIIS Staff can add in the keyword list. The keyword entity includes keywordID (a unique identifier) and keywordName (must also be unique), representing the topic associated with a research entry. They can also link and unlink keywords with research entries.

Panel

MCIIS Staff can add, edit, and delete the list of faculty panelists to research projects. A faculty member can be a panelist for multiple research projects, while a research a join table that handles the many-to-many relationship between faculty members and research entries that includes panelID (a unique identifier), facultyID (foreign key referencing the Faculty entity) and researchID (foreign key referencing the Research entity).

Research Entry Log

MCIIS Staff creation, modification, and archiving of a research entry is tracked through the research entry log. It tracks all actions related to research entries, including direct modifications to research attributes and any changes to associated entities (keywords, researchers, panelists, SDG, SRIG, and Agenda). The Research Entry Log entity includes entryLogID (a unique identifier), performedBy (referencing the User entity), actionType (create or modify), researchID (referencing the Research entity), and timestamp (indicating the date and time of the upload).

Agenda

MCIIS Staff can categorize and align research outputs with institutional or college research priorities to support thematic analysis and strategic reporting. The Agenda entity provides a structured way to classify research according to established research agendas. It includes the following attributes: agendaID (a unique identifier), agendaName (the name of the research agenda, e.g., “University Research Agenda”), and agendaDescription (a text field that provides details on the scope, objectives, or description of the agenda).

SDG

MCIIS Staff can align research projects with the United Nations Sustainable Development Goals (SDGs) to support institutional reporting and global agenda alignment. The SDG entity represents these goals and their related descriptions. It includes the following attributes: sdgID (a unique identifier), sdgName (the name of the SDG, e.g., “SDG4 – Quality Education”), and sdgDescription (a text field containing the official UN or localized context of the SDG).

SRIG

MCIIS Staff can categorize research outputs under thematic clusters to better organize and analyze research trends within the institution. The SRIG (Strategic Research Interest Group) entity defines specific research clusters or focus areas. It includes the following attributes: srigID (a unique identifier), srigName (the name of the research interest group, e.g., “Artificial

Intelligence & Machine Learning”), and srigDescription (a text field explaining the research themes and topics that fall under this SRIG).

Transaction Requirements:

Data Entry

- Add a new research entry with complete details.
- Add a new researcher and associate them with a research entry.
- Add a new keyword to the Keyword table (if it does not already exist, if it exists, simply add a record in the ResearchKeyword join table).
- Add a record in the ResearchKeyword join table to associate a keyword to a research entry.
- Add a record in the Panelist join table to associate a faculty to a research entry.
- Add a record in the ResearchAgenda, ResearchSRIG, and ResearchSDG join table to associate a research under Agenda, SDG, or SRIG.

Data Update/Deletion

- Update the details of an existing research entry.
- Update the information of a researcher.
- Update the associated Agenda, SDG, or SRIG of a research.
- Update a record in the ResearchAgenda, ResearchSRIG, and ResearchSDG join table to associate a research under Agenda, SDG, or SRIG.
- Delete a researcher associated with a research.
- Delete a record in the ResearchKeyword join table to unlink a keyword from a research entry.
- Delete a record in the Panelist join table to unlink a faculty from a research entry.
- Delete the associated Agenda, SDG, or SRIG of a research.

Data Queries

- Search research by title or keyword.
- Filter research by adviser, program, or year.
- View profile information of faculty members.
- Generate a list of all faculty with their advised and paneled research counts.
- Generate a report showing the number and percentage of research per program are associated with Agenda, SRIG, and SDG.
- Generate a report showing the number and percentage of research per year within the program are associated with Agenda, SRIG, and SDG.
- Identify the top 5 research advisers who have the most number of advised research projects.

- Identify the top 5 research panelists who have the most number of paneled research projects.
- View the full information of a research including researchID, researchTitle, program, publishedMonth, publishedYear, researchAbstract, researchAdviser, researchers, panelists, keywords, researchApprovalSheet, and researchManuscript.

Faculty User Views of MCIS Repository

Data Requirements:

Research

Faculty members can add and edit research entries they advise on, ensuring that all research metadata is accurate and properly documented. The Research entity includes researchID (unique identifier), uploadedBy (referencing the User entity), researchTitle (must be unique), researchAdviser (referencing the Faculty entity), program (“Bachelor of Science in Information Technology,” “Bachelor of Science in Computer Science,” “Bachelor of Library and Information Science,” “Master of Library and Information Science,” or “Master in Information Technology”), publishedMonth, publishedYear, researchAbstract (text), researchApprovalSheet (image), and researchManuscript (pdf file). They can also filter it by researchAdviser, program, and year.

Thematic tagging: The research entry they advised can be associated with one or more Agenda, SDG, and SRIG categories to enable classification, reporting, and dashboard analytics.

Faculty

Faculty members can view the faculty list and edit their own faculty information. The Faculty entity includes facultyID (unique identifier), firstName, middleName, lastName, position, designation, email (valid USeP email and must be unique), ORCID, contactNumber, educationalAttainment, fieldOfSpecialization, and researchInterest.

Researcher

Faculty members can add researcher information and assign them to the research they participated in. The Researcher entity includes researcherID, researchID (referencing the Research entity), firstName, middleName, lastName, and email (valid USeP email and must be unique).

Keyword

Faculty members can create new keyword records if they do not exist and associate them with research entries but cannot modify or delete keywords. The keyword entity includes keywordID (a unique identifier) and keywordName (must also be unique).

Panel

Faculty members can view the list of faculty panelists to research projects. A faculty member can be a panelist for multiple research projects, while a research project can have multiple panelists. The Panel entity is a join table that handles the many-to-many relationship between faculty members and research entries that includes panelID (a unique identifier), facultyID (foreign key referencing the Faculty entity) and researchID (foreign key referencing the Research entity).

Research Entry Log

Faculty's creation, modification, and archiving of a research entry they advised is tracked through the research entry log. It tracks all actions related to research entries, including direct modifications to research attributes and any changes to associated entities (keywords, researchers, panelists, SDG, SRIG, and Agenda). The Research Entry Log entity includes entryLogID (a unique identifier), performedBy (referencing the User entity), actionType (create or modify), researchID (referencing the Research entity), and timestamp (indicating the date and time of the upload).

Agenda

Faculty can categorize and align the research entry they advised with institutional or college research priorities to support thematic analysis and strategic reporting. The Agenda entity provides a structured way to classify research according to established research agendas. It includes the following attributes: agendaID (a unique identifier), agendaName (the name of the research agenda, e.g., "University Research Agenda"), and agendaDescription (a text field that provides details on the scope, objectives, or description of the agenda).

SDG

Faculty can align the research entry they advised with the United Nations Sustainable Development Goals (SDGs) to support institutional reporting and global agenda alignment. The SDG entity represents these goals and their related descriptions. It includes the following attributes: sdgID (a unique identifier), sdgName (the name of the SDG, e.g., "SDG4 – Quality Education"), and sdgDescription (a text field containing the official UN or localized context of the SDG).

SRIG

Faculty can categorize the research entry they advised research outputs under thematic clusters to better organize and analyze research trends within the institution. The SRIG (Strategic Research Interest Group) entity defines specific research clusters or focus areas. It includes the

following attributes: srigID (a unique identifier), srigName (the name of the research interest group, e.g., “Artificial Intelligence & Machine Learning”), and srigDescription (a text field explaining the research themes and topics that fall under this SRIG).

Transaction Requirements:

Data Entry

- Add a new research entry with complete details.
- Add a new researcher and associate them with a research entry.
- Add a new keyword to the Keyword table (if it does not already exist, if it exists, simply add a record in the ResearchKeyword join table).
- Add a record in the ResearchKeyword join table to associate a keyword to a research entry.
- Add a record in the Panelist join table to associate a faculty to a research entry.
- Add a record in the ResearchAgenda, ResearchSRIG, and ResearchSDG join table to associate a research under Agenda, SDG, or SRIG.

Data Update/Deletion

- Update the details of their own faculty information.
- Update the details of a research entry they advised.
- Update the information of a researcher of the research they advised.
- Update a record in the ResearchAgenda, ResearchSRIG, and ResearchSDG join table to associate a research under Agenda, SDG, or SRIG.
- Delete a researcher of the research they advised.
- Delete a record in the ResearchKeyword join table to unlink a keyword from a research entry of the research they advised.
- Delete a record in the Panelist join table to unlink a faculty from a research entry.

Data Queries

- Search research by title or keyword.
- Filter research by adviser, program, or year.
- View profile information of faculty members.
- Generate a report showing the number of research entries they have advised or paneled.
- View the full information of a research including researchID, researchTitle, program, publishedMonth, publishedYear, researchAbstract, researchAdviser, researchers, panelists, keywords, researchApprovalSheet, and researchManuscript.

Student User Views of MCIS Repository

Data Requirements:

Research

Students can edit existing research entries where they are the researcher (link through researcher and user email). The research entry must be precreated by the faculty who is the adviser of that research. The Research entity includes researchID (unique identifier), uploadedBy (referencing the User entity), researchTitle (must be unique), researchAdviser (referencing the Faculty entity), program (“Bachelor of Science in Information Technology,” “Bachelor of Science in Computer Science,” “Bachelor of Library and Information Science,” “Master of Library and Information Science,” or “Master in Information Technology”), publishedMonth, publishedYear, researchAbstract (text), researchApprovalSheet (image), and researchManuscript (pdf file). They can also filter it by researchAdviser, researchPanel, researcher, publishedDate, program, and keyword.

Thematic tagging: The research entry where they are the researcher can be associated with one or more Agenda, SDG, and SRIG categories to enable classification, reporting, and dashboard analytics.

Faculty

Students can view information about faculty members. The Faculty entity includes facultyID (unique identifier), firstName, middleName, lastName, position, designation, email (valid USEP email and must be unique), ORCID, contactNumber, educationalAttainment, fieldOfSpecialization, and researchInterest.

Researcher

Students can view the list of researchers. The Researcher entity includes researcherID, researchID (referencing the Research entity), firstName, middleName, lastName, and email (valid USEP email and must be unique).

Keyword

Students can search research using keywords. The keyword entity includes keywordID (a unique identifier) and keywordName (must also be unique).

Panel

Students can view the list of faculty panelists to research projects. A faculty member can be a panelist for multiple research projects, while a research project can have multiple panelists. The

Panel entity is a join table that handles the many-to-many relationship between faculty members and research entries that includes panelID (a unique identifier), facultyID (foreign key referencing the Faculty entity), and researchID (foreign key referencing the Research entity).

Agenda

Students can categorize and align the research entry where they are the researcher with institutional or college research priorities to support thematic analysis and strategic reporting. The Agenda entity provides a structured way to classify research according to established research agendas. It includes the following attributes: agendaID (a unique identifier), agendaName (the name of the research agenda, e.g., “University Research Agenda”), and agendaDescription (a text field that provides details on the scope, objectives, or description of the agenda).

SDG

Students can align the research entry where they are the researcher with the United Nations Sustainable Development Goals (SDGs) to support institutional reporting and global agenda alignment. The SDG entity represents these goals and their related descriptions. It includes the following attributes: sdgID (a unique identifier), sdgName (the name of the SDG, e.g., “SDG4 – Quality Education”), and sdgDescription (a text field containing the official UN or localized context of the SDG).

SRIG

Students can categorize the research entry where they are the researcher under thematic clusters to better organize and analyze research trends within the institution. The SRIG (Strategic Research Interest Group) entity defines specific research clusters or focus areas. It includes the following attributes: srigID (a unique identifier), srigName (the name of the research interest group, e.g., “Artificial Intelligence & Machine Learning”), and srigDescription (a text field explaining the research themes and topics that fall under this SRIG).

Transaction Requirements:

Data Entry

- Add a new keyword to the Keyword table (if it does not already exist, if it exists, simply add a record in the ResearchKeyword join table).

- Add a record in the ResearchKeyword join table to associate a keyword to a research entry.
- Add a record in the Panelist join table to associate a faculty to a research entry.
- Add a record in the ResearchAgenda, ResearchSRIG, and ResearchSDG join table to associate a research under Agenda, SDG, or SRIG.

Data Update/Deletion

- Update the details of a research entry where they are the researcher.
- Update a record in the ResearchAgenda, ResearchSRIG, and ResearchSDG join table to associate a research under Agenda, SDG, or SRIG.
- Delete a record in the ResearchKeyword join table to unlink a keyword from a research entry of the research they advised.
- Delete a record in the Panelist join table to unlink a faculty from a research entry.

Data Queries

- Search research by title or keyword.
- Filter research by adviser, program, or year.
- View profile information of faculty members.
- View the full information of a research including researchID, researchTitle, program, publishedMonth, publishedYear, researchAbstract, researchAdviser, researchers, panelists, keywords, researchApprovalSheet, and researchManuscript.